CSE400 Lecture 20 Scribe

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Imagine you are expert scribe generator. Make a lecture scribe suitable for exam preparation for CSE400 Lecture 20: Linear Transformations and Expectations of Multiple RVs. When you are writing the scribe, do reasoning step by step through definitions, assumptions, proofs, derivations, and worked example solutions exactly as presented in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. Use only the provided lecture slides and relevant textbook excerpts. Don't mention any material, explanations or examples that are not present in the context.
Organize the scribe clearly in a formal academic style.

Here is a **formal, exam-oriented lecture scribe** for **CSE400 Lecture 20: Linear Transformations and Expectations of Multiple Random Variables**, strictly based on the provided slides.

CSE400 Lecture 20 Scribe

Linear Transformations and Expectations of Multiple Random Variables

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1. Random Vector Representation

Definition

For multiple random variables $X_1, X_2, \ldots, X_N$, we represent them in **vector form**:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- This vector $\mathbf{X}$ is called a **random vector**.
- It provides a **compact representation** of multiple random variables.

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2. Mean Vector (Expectation of a Random Vector)

Definition

The mean vector of a random vector $\mathbf{X}$ is:

$$\boldsymbol{\mu}_X = \mathbb{E}[\mathbf{X}]$$

Interpretation

- Represents the **average value** of the random vector.
- Each element corresponds to:

$$\boldsymbol{\mu}_X = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_N] \end{bmatrix}$$

---

3. Correlation Matrix

Definition

The correlation matrix of a random vector $\mathbf{X}$ is:

$$R_{XX} = \mathbb{E}[\mathbf{X}\mathbf{X}^T]$$

Element-wise Representation

$$R_{XX}(i,j) = \mathbb{E}[X_i X_j]$$

Matrix Form

$$R_{XX} = \begin{bmatrix} \mathbb{E}[X_1X_1] & \mathbb{E}[X_1X_2] & \cdots \\ \mathbb{E}[X_2X_1] & \mathbb{E}[X_2X_2] & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property

- R_{XX} is a **symmetric matrix**:

$$R_{XX}(i,j) = R_{XX}(j,i)$$

---

4. Covariance Matrix

Definition

For a random vector $\mathbf{X}$, the covariance matrix is:

$$C_{XX} = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T]$$

Structure (Interpretation)

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property

- Symmetry:

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Thus, covariance matrix acts as a "**relationship map" of dependencies between variables**.

---

5. Linear Transformation of Random Vectors

Model

Consider a linear transformation:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Where:

- A: Transformation matrix (scaling/rotation)
- $\mathbf{b}$: Constant vector (shift/translation)
- $\mathbf{X}$: Input random vector

Component-wise Representation

$$Y_i = \sum_{j=1}^{N} a_{ij} X_j + b_i$$

---

6. Mean of Transformed Random Vector

Goal

Find:

$$\boldsymbol{\mu}_Y = \mathbb{E}[\mathbf{Y}]$$

Step-by-Step Derivation

Substitute:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

$$\boldsymbol{\mu}_Y = \mathbb{E}[A\mathbf{X} + \mathbf{b}]$$

Apply linearity of expectation:

$$\boldsymbol{\mu}_Y = \mathbb{E}[A\mathbf{X}] + \mathbb{E}[\mathbf{b}]$$

Since:

- A is constant
- $\mathbf{b}$ is constant

$$\mathbb{E}[A\mathbf{X}] = A\mathbb{E}[\mathbf{X}] = A\boldsymbol{\mu}_X$$

$$\mathbb{E}[\mathbf{b}] = \mathbf{b}$$

Final Result

$$\boxed{\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + \mathbf{b}}$$

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7. Correlation Matrix Transformation

Definition

$$R_{YY} = \mathbb{E}[\mathbf{Y}\mathbf{Y}^T]$$

Substitute:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Expansion

$$R_{YY} = \mathbb{E}[(A\mathbf{X} + \mathbf{b})(A\mathbf{X} + \mathbf{b})^T]$$
$$= \mathbb{E}[A\mathbf{X}\mathbf{X}^T A^T + A\mathbf{X}\mathbf{b}^T + \mathbf{b}\mathbf{X}^T A^T + \mathbf{b}\mathbf{b}^T]$$

Apply Expectation

$$R_{YY} = A\mathbb{E}[\mathbf{X}\mathbf{X}^T]A^T + A\mathbb{E}[\mathbf{X}]\mathbf{b}^T + \mathbf{b}\mathbb{E}[\mathbf{X}]^T A^T + \mathbf{b}\mathbf{b}^T$$

Final Result

$$\boxed{R_{YY} = AR_{XX}A^T + A\boldsymbol{\mu}_X\mathbf{b}^T + \mathbf{b}\boldsymbol{\mu}_X^T A^T + \mathbf{b}\mathbf{b}^T}$$

---

8. Covariance Matrix Transformation

Key Observation

$$\mathbf{Y} - \boldsymbol{\mu}_Y = A(\mathbf{X} - \boldsymbol{\mu}_X)$$

- The **shift vector** $\mathbf{b}$ cancels out
- Covariance depends only on deviations from mean

Derivation

$$C_{YY} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^T]$$

Substitute:

$$= \mathbb{E}[A(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T A^T]$$

Final Result

$$\boxed{C_{YY} = AC_{XX}A^T}$$

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9. Case Study: Smart City Traffic + Weather System

Setup

- Model traffic congestion using multiple variables (e.g., rain, speed, pollution, temperature).
- Represent system as an N-dimensional random vector.

Steps

1. **Define Random Vector**
   - Combine all variables into one vector.
2. **Mean Vector**
   - Represents average system behavior.
3. **Covariance Matrix**
   - Captures dependencies:
     - Rain ↑ ⇒ Traffic ↑
     - Rain ↑ ⇒ Speed ↓
     - Traffic ↑ ⇒ Pollution ↑
     - Temperature ↑ ⇒ Pollution disperses
4. **Interpretation**
   - Covariance matrix acts as a **dependency map of the system**.

Key Takeaways (Exam Focus)

- Random variables → **Random Vector**
- Expectation → **Mean Vector**
- $R_{XX} = \mathbb{E}[XX^T]$ (Correlation)

- $C_{XX} = \mathbb{E}[(X-\mu)(X-\mu)^T]$ (Covariance)
- Linear transformation:

$$Y = AX + b$$

- Mean transformation:

$$\mu_Y = A\mu_X + b$$

- Covariance transformation:

$$C_{YY} = AC_{XX}A^T$$

- Shift b **does not affect covariance**

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Here is a **formal, exam-oriented lecture scribe** for:

CSE400 Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

(Based strictly on provided lecture slides)

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1. Introduction and Scope

This lecture focuses on extending probability concepts from single random variables to **multiple random variables**, represented compactly using vectors. The key objectives include:

- Statistical representation of random vectors
- Mean, correlation, and covariance matrices
- Linear transformations of random vectors
- Mapping properties under transformations of the form $Y = AX + b$

---

2. Random Vector Representation

Definition

For multiple random variables $X_1, X_2, \ldots, X_N$, we define:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Interpretation

- $\mathbf{X}$ is called a **random vector**.
- It provides a **compact representation of multiple random variables**.

(As shown in the diagram on page 3, where individual RVs are stacked into a column vector.)

---

3. Mean Vector (Expectation of a Random Vector)

Definition

$$\boldsymbol{\mu}_X = \mathbb{E}[\mathbf{X}]$$

Structure

$$\boldsymbol{\mu}_X = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_N] \end{bmatrix}$$

Interpretation

- Represents the **average value of the random vector**
- Each component corresponds to the mean of one variable

(From page 4–5, where the expectation is applied element-wise.)

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4. Correlation Matrix

Definition

$$R_{XX} = \mathbb{E}[\mathbf{X}\mathbf{X}^T]$$

Element-wise Form

$$R_{XX}(i,j) = \mathbb{E}[X_i X_j]$$

Matrix Representation

$$R_{XX} = \begin{bmatrix} \mathbb{E}[X_1X_1] & \mathbb{E}[X_1X_2] & \cdots \\ \mathbb{E}[X_2X_1] & \mathbb{E}[X_2X_2] & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property

- R_{XX} is **symmetric**, since:

$$\mathbb{E}[X_iX_j] = \mathbb{E}[X_jX_i]$$

(Explicitly stated on page 7.)

---

5. Covariance Matrix

Definition

$$C_{XX} = \mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T\right]$$

Expanded Structure

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property

- Symmetry:

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Interpretation

- Acts as a **relationship map of random variables**
- Diagonal → variances
- Off-diagonal → pairwise dependencies

(Explained with matrix structure and symmetry on pages 8–9.)

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6. Linear Transformation of Random Vectors

Model Definition

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Where:

- A: transformation matrix (scales/rotates data)
- $\mathbf{b}$: constant vector (shifts/translates center)
- $\mathbf{X}$: input random vector

(Described on page 10.)

Component-wise Form

$$Y_i = \sum_{j=1}^{N} a_{ij} X_j + b_i$$

---

7. Mean of Transformed Random Vector

Objective

$$\boldsymbol{\mu}_Y = \mathbb{E}[\mathbf{Y}]$$

Step-by-Step Derivation

Step 1: Substitute transformation

$$\boldsymbol{\mu}_Y = \mathbb{E}[A\mathbf{X} + \mathbf{b}]$$

Step 2: Apply linearity of expectation

$$\boldsymbol{\mu}_Y = \mathbb{E}[A\mathbf{X}] + \mathbb{E}[\mathbf{b}]$$

Step 3: Use constants property

- A is constant ⇒ $\mathbb{E}[A\mathbf{X}] = A\mathbb{E}[\mathbf{X}]$
- $\mathbf{b}$ is constant ⇒ $\mathbb{E}[\mathbf{b}] = \mathbf{b}$

Final Result

$$\boxed{\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + \mathbf{b}}$$

(Derived explicitly across pages 12–13.)

---

8. Correlation Matrix Transformation

Definition

$$R_{YY} = \mathbb{E}[\mathbf{Y}\mathbf{Y}^T]$$

Step-by-Step Expansion

Step 1: Substitute $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$

$$R_{YY} = \mathbb{E}[(A\mathbf{X} + \mathbf{b})(A\mathbf{X} + \mathbf{b})^T]$$

Step 2: Expand terms

$$R_{YY} = \mathbb{E}[A\mathbf{X}\mathbf{X}^T A^T + A\mathbf{X}\mathbf{b}^T + \mathbf{b}\mathbf{X}^T A^T + \mathbf{b}\mathbf{b}^T]$$

Step 3: Apply expectation

$$R_{YY} = A\mathbb{E}[\mathbf{X}\mathbf{X}^T]A^T + A\mathbb{E}[\mathbf{X}]\mathbf{b}^T + \mathbf{b}\mathbb{E}[\mathbf{X}]^T A^T + \mathbf{b}\mathbf{b}^T$$

Final Result

$$\boxed{R_{YY} = AR_{XX}A^T + A\boldsymbol{\mu}_X\mathbf{b}^T + \mathbf{b}\boldsymbol{\mu}_X^T A^T + \mathbf{b}\mathbf{b}^T}$$

(Derived step-by-step on pages 14–17.)

---

9. Covariance Matrix Transformation

Key Observation

From page 18:

$$\mathbf{Y} - \boldsymbol{\mu}_Y = A(\mathbf{X} - \boldsymbol{\mu}_X)$$

- The shift $\mathbf{b}$ cancels out
- Covariance depends only on deviations from the mean

Derivation

$$C_{YY} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^T]$$

Substitute:

$$= \mathbb{E}[A(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T A^T]$$

Take constants out:

$$= A\mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T]A^T$$

Final Result

$$\boxed{C_{YY} = AC_{XX}A^T}$$

(Based on the key observation on page 18.)

---

10. Case Study: Smart City Traffic + Weather System

(Pages 53–57)

Step 1: Define Random Variables

- Multiple variables (traffic, weather, etc.)
- Combined into an N-dimensional random vector

Step 2: Mean Vector

- Represents average system behavior

Step 3: Covariance Matrix (Interpretation)

From the slide:

- More rain ⇒ more congestion
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

Conclusion

- Covariance matrix acts as a **dependency map of the city system**

Final Exam Summary

- Random variables → **Random Vector**
- Mean:

$$\mu_X = \mathbb{E}[X]$$

- Correlation:

$$R_{XX} = \mathbb{E}[XX^T]$$

- Covariance:

$$C_{XX} = \mathbb{E}[(X-\mu)(X-\mu)^T]$$

- Linear transformation:

$$Y = AX + b$$

- Mean transformation:

$$\mu_Y = A\mu_X + b$$

- Correlation transformation:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

- Covariance transformation:

$$C_{YY} = AC_{XX}A^T$$

- **Important Insight**: Shift b does **not affect covariance**

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Here is a **complete, formal, exam-oriented lecture scribe** for:

CSE400 Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

(Strictly based on lecture slides)

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1. Introduction and Scope

This lecture develops a structured framework for handling **multiple random variables** using vector and matrix representations. The main objectives include:

- Statistical representation of random vectors
- Mean vector, correlation matrix, covariance matrix
- Linear transformations of random vectors
- Mapping properties under $Y = AX + b$
- Practical interpretation using real-world systems

(As outlined on page 2 of the slides.)

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2. Random Vector Representation

Definition

For random variables $X_1, X_2, \ldots, X_N$, define:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Interpretation

- $\mathbf{X}$ is called a **random vector**
- It provides a **compact representation of multiple random variables**

(Clearly illustrated in the diagram on page 3.)

---

3. Mean Vector (Expectation of Random Vector)

Definition

$$\boldsymbol{\mu}_X = \mathbb{E}[\mathbf{X}]$$

Structure

$$\boldsymbol{\mu}_X = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_N] \end{bmatrix}$$

Interpretation

- Represents the **average value of the random vector**
- Each element corresponds to the mean of a component variable

(From pages 4–5, including the example structure.)

---

4. Correlation Matrix

Definition

$$R_{XX} = \mathbb{E}[\mathbf{X}\mathbf{X}^T]$$

Element-wise Form

$$R_{XX}(i,j) = \mathbb{E}[X_i X_j], \quad i,j \leq N$$

Matrix Representation

$$R_{XX} = \begin{bmatrix} \mathbb{E}[X_1X_1] & \mathbb{E}[X_1X_2] & \cdots \\ \mathbb{E}[X_2X_1] & \mathbb{E}[X_2X_2] & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property

$$R_{XX} \text{ is symmetric}$$

Reasoning:

$$\mathbb{E}[X_i X_j] = \mathbb{E}[X_j X_i]$$

(Derived and visualized on pages 6–7.)

---

5. Covariance Matrix

Definition

$$C_{XX} = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T]$$

Expanded Form

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Property (Symmetry)

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Interpretation

- Diagonal elements → variances
- Off-diagonal elements → dependencies
- Acts as a **relationship map of random variables**

(Explained with visual matrix on pages 8–9.)

---

6. Linear Transformation of Random Vectors

Model Definition

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Components

- A: transformation matrix (scales/rotates data)
- $\mathbf{b}$: constant vector (shifts/translates center)
- $\mathbf{X}$: input random vector

(Given on page 10.)

Component-wise Expansion

$$Y_i = \sum_{j=1}^{N} a_{ij} X_j + b_i$$

---

7. Mean of Transformed Random Vector

Objective

$$\boldsymbol{\mu}_Y = \mathbb{E}[\mathbf{Y}]$$

Step-by-Step Derivation

Step 1: Substitute transformation

$$\boldsymbol{\mu}_Y = \mathbb{E}[A\mathbf{X} + \mathbf{b}]$$

Step 2: Apply linearity of expectation

$$\boldsymbol{\mu}_Y = \mathbb{E}[A\mathbf{X}] + \mathbb{E}[\mathbf{b}]$$

Step 3: Use constant properties

- A is constant ⇒ $\mathbb{E}[A\mathbf{X}] = A\mathbb{E}[\mathbf{X}]$
- $\mathbf{b}$ is constant ⇒ $\mathbb{E}[\mathbf{b}] = \mathbf{b}$

Final Result

$$\boxed{\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + \mathbf{b}}$$

(Derived explicitly on pages 11–13.)

---

8. Correlation Matrix Transformation

Definition

$$R_{YY} = \mathbb{E}[\mathbf{Y}\mathbf{Y}^T]$$

Step-by-Step Derivation

Step 1: Substitute $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$

$$R_{YY} = \mathbb{E}[(A\mathbf{X} + \mathbf{b})(A\mathbf{X} + \mathbf{b})^T]$$

Step 2: Expand product

$$R_{YY} = \mathbb{E}[A\mathbf{X}\mathbf{X}^T A^T + A\mathbf{X}\mathbf{b}^T + \mathbf{b}\mathbf{X}^T A^T + \mathbf{b}\mathbf{b}^T]$$

Step 3: Apply expectation and constants

$$R_{YY} = A\mathbb{E}[\mathbf{X}\mathbf{X}^T]A^T + A\mathbb{E}[\mathbf{X}]\mathbf{b}^T + \mathbf{b}\mathbb{E}[\mathbf{X}]^T A^T + \mathbf{b}\mathbf{b}^T$$

Final Result

$$\boxed{R_{YY} = AR_{XX}A^T + A\boldsymbol{\mu}_X\mathbf{b}^T + \mathbf{b}\boldsymbol{\mu}_X^T A^T + \mathbf{b}\mathbf{b}^T}$$

(Derived across pages 14–17.)

---

9. Covariance Matrix Transformation

Key Observation

$$\mathbf{Y} - \boldsymbol{\mu}_Y = A(\mathbf{X} - \boldsymbol{\mu}_X)$$

- The shift $\mathbf{b}$ cancels out
- Covariance depends only on deviation from mean

(Explicitly stated on page 18.)

Derivation

$$C_{YY} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^T]$$

Substitute:

$$= \mathbb{E}[A(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T A^T]$$

Take constants out:

$$= A\mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T]A^T$$

Final Result

$$\boxed{C_{YY} = AC_{XX}A^T}$$

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10. Case Study: Smart City Traffic + Weather System

(Pages 53–57)

Step 1: Define Random Variables

- Multiple variables (traffic, rain, temperature, pollution, etc.)
- Represented as an N-dimensional random vector

Step 2: Mean Vector

- Captures average system behavior

Step 3: Covariance Matrix Interpretation

From slide interpretation:

- More rain ⇒ more congestion
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

Conclusion

- Covariance matrix acts as a "**dependency map of the city**"

Final Exam Summary (Must Remember)

Core Definitions

$$\mu_X = \mathbb{E}[X], \quad R_{XX} = \mathbb{E}[XX^T], \quad C_{XX} = \mathbb{E}[(X-\mu)(X-\mu)^T]$$

Linear Transformation

$$Y = AX + b$$

Key Results

- Mean:

$$\mu_Y = A\mu_X + b$$

- Correlation:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

- Covariance:

$$C_{YY} = AC_{XX}A^T$$

Critical Insight

- **Covariance is invariant to shifts** (independent of b)
- Only affected by linear transformation A

If you want, I can now convert this into a **clean LaTeX code (exam-ready PDF like your previous scribes)**.